

B M Tazbiul Hassan Anik, Ph.D.

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PROFESSIONAL INTERESTS

Active Traffic Management (ATM), Safety & Operations, Crash & Conflict Data Analytics, Signal Optimization, Uncertainty Quantification, Causality, Predictive & Preventive Modeling, Simulation

EDUCATION

Ph.D. in Transportation Systems Engineering (CGPA: 3.94 / 4.00)	June 2025
University of Central Florida	Orlando, FL
B.Sc. in Civil Engineering (Major: Transportation)	February 2021
Bangladesh University of Engineering and Technology	Dhaka, Bangladesh

PROFESSIONAL EXPERIENCE

Graduate Research Assistant	August 2022 – Present
University of Central Florida	Orlando, FL
Traffic Data Analyst	July 2021 – June 2022
Accident Research Institute	Dhaka, Bangladesh

TECHNICAL SKILLS

Programming Languages:	Python, R, SQL, HTML5, CSS, JavaScript
Libraries & Frameworks:	Node.js, Express.js, React, Pandas, NumPy, Scikit-learn, PyTorch
AI & ML:	Transformer, Bayesian Modeling, Regression Modeling, Extreme Value Theory, Causal Forest, DiD, A/B Testing
Transportation Modeling:	VISSIM, Synchro, HCS, Cube
GIS and Mapping Software:	ArcGIS, QGIS, Google Earth Pro
Data Engineering & Pipelines:	MongoDB, ETL Pipeline Design, Data Warehousing, CI/CD
Data Visualization & Reporting:	D3.js, Tableau, Matplotlib, Seaborn, Plotly
Development & Collaboration:	Git, GitHub, Docker, Agile

SELECTED PROJECTS

‘Prediction of Safety Opportunities Using Automated Traffic Signal Performance Measures’ — FDOT

- Created a **Python Framework** (‘citysignal’) to automatically transform high-resolution ATSPM **controller event logs** into signal performance measures (e.g., occupancy, split failure, pedestrian delay, etc.) across intersections.
- Applied **Bayesian Beta-Binomial models** to estimate interval-level pedestrian arrival probabilities, capturing **posterior uncertainty** and improving signal control strategies under sparse data.
- Implemented a scoring framework to **rank intersections based on safety-critical performance measures**, enabling prioritization of interventions across a large signal network.
- Developed and evaluated data-driven interventions such as Leading Pedestrian Interval, No Turn on Red, and Pedestrian Recall, validating a **10% reduction in conflicts** and **25% decrease in pedestrian delay**.
- Built an [interactive dashboard](#) (MongoDB, Node.js/Express.js, React) to visualize performance measures and monitor safety & operation interventions, **to be integrated within FDOT’s ATMS**.

‘After Deployment Evaluation for ATTAIN’ — FDOT

- Conducted **before-and-after performance evaluation of TSM&O strategies** using crash data (S4A), traffic data, and survey responses to assess impacts on safety and mobility.
- Created GIS-based basemaps for Florida’s I-4 and adjacent arterials (ICM corridors), **integrating geospatial and sensor data to evaluate travel time along detours**.
- Analyzed UCF shuttle GPS data and on-campus parking behavior to develop **mobility strategies enhancing multimodal access, optimizing parking demand**.
- Performed **Benefit-Cost Analysis (BCA)** and analyzed **improvements in travel time reliability**, UCF shuttle **headways (TSP)**, **parking behavior**, and **mode choice** influenced by real-time information systems (e.g., FL511).

‘Intersection Safety and Mobility Analytics Using Sensor Fusion’ — FDOT

- Leading the integration connected vehicle telemetry (OBU/GPS/IMU) and infrastructure sensors (CCTV, radar) at signalized intersections to enable robust and real-time safety perception.

- Designing perception pipelines for **early detection of unsafe interactions** using surrogate safety measures (e.g., PET, TTC), enhancing **AV situational awareness at intersections**.
- Derived **driving behavior profiles** (e.g., harsh braking, high deceleration, speed spikes) to flag aggressive maneuvers and correlate with near-miss events.

‘Phase IV - ATTAIn: Expanding the Real-Time Safety System to include I-4 Ultimate, I-75 and I-95, Conflict Analysis and System Migration’ — FDOT

- Developed a **real-time crash prediction system** (of primary and secondary crash risk) integrating multi-source data (traffic, weather, and crash reports) using advanced machine learning models (Transformer, LSTM, CNN).
- Defined and evaluated key performance indicators such as sensitivity and false alarm rates (FAR) across **segment types** (**basic, merge, weaving, express lanes**) and **crash severity** (**KABCO scale**).

‘After Study Evaluation of Interstate 4 (I-4) Florida’s Regional Advanced Mobility Elements (FRAME) Project’ — FDOT

- **Led the evaluation of safety and mobility impacts** during the deployment of ITS and Connected Vehicle (CV) technologies along I-4 and adjacent arterials, using multi-source data (S4A, MVDS, INRIX, and weather).
- Analyzed **crash trends** and **freeway performance metrics** such as Planning Time Index (PTI), travel time reliability, and delay to assess effectiveness of Dynamic Route Guidance, Incident Management systems, etc.

REVIEW EXPERIENCE

Conference:	Transportation Research Board (TRB)
Journal:	Engineering Applications of Artificial Intelligence, Accident Analysis & Prevention, Transportation Research Part C: Emerging Technologies, Heliyon

PUBLICATIONS

- **Anik, BM Tazbiul Hassan**, Zubayer Islam, and Mohamed Abdel-Aty. “A time-embedded attention-based transformer for crash likelihood prediction at intersections using connected vehicle data.” Transportation Research Part C: Emerging Technologies 169 (2024): 104831.
- Islam, Zubayer, Mohamed Abdel-Aty, and **BM Tazbiul Hassan Anik**. “Transformer-conformer ensemble for crash prediction using connected vehicle trajectory data.” IEEE Journal of Intelligent Transportation Systems 4 (2023): 979-988.
- **Anik, BM Tazbiul Hassan**, Mohamed Abdel-Aty, and Zubayer Islam. “Can we realize seamless traffic safety at smart intersections by predicting and preventing impending crashes?.” Accident Analysis & Prevention 211 (2025): 107908.
- Raihan, Md Asif, **BM Tazbiul Hassan Anik**, Fajle Rabbi Ashik, Md Mahmud Hasan, and S. M. Mahmud. “Motorcycle helmet use behavior: What does the data tell us?”. No. TRBAM-23-03573. 2023.
- Abdel-Aty, Mohamed, Tarek Hasan, and **BM Tazbiul Hassan Anik**. “An advanced real-time crash prediction framework for hard shoulder running and variable speed limits system using transformer.” Scientific Reports 14.1 (2024): 26403.
- **Anik, BM Tazbiul Hassan**. “Identifying Factors Contributing to Pedestrian Crash Injury Severity: A Logit Modeling Approach.”
- Rabbani, Borhan Uddin, and **BM Tazbiul Hassan Anik**. “A Comparative Study on Machine Learning Algorithms for Crash Severity Prediction: A Case Study for Bangladesh.” Available at SSRN 4611094 (2023).
- Oishi, Azka Rodoshi, Md Jamil Ahsan, Azka Sejuti, and **Anik, BM Tazbiul Hassan**. “Evaluating Traumatic Brain Injury Outcomes in Road Traffic Crashes.” arXiv preprint arXiv:2411.01771 (2024).
- **Anik, BM Tazbiul Hassan**, Tinvi Islam Tisha, Md Shamim Iftakar, Md Asif Raihan, Md Hadiuzzaman. “Factors Contributing to Heavy Vehicle Crash Severity in Bangladesh: A Multinomial Logit Modeling Approach.”